

UNIT – 5
ASSESSMENT – 1
OUTPUT

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USER-ITEM INTERACTIONS

('U1', 'I1', 5)

('U1', 'I2', 4)

('U1', 'I3', 2)

...

CONTENT-BASED RECOMMENDATIONS

['I1', 'I2', 'I4']

SIMILAR USERS

['U2', 'U3', 'U4']

COLLABORATIVE RECOMMENDATIONS

['I4', 'I5']

FINAL HYBRID RECOMMENDATIONS

I1

I2

I4

I5

INTELLIGENT AGENT

Agent detected positive feedback.

1. Agent increases priority of similar items.

```
('ZoneA', 'Flood', 8)
('ZoneB', 'Earthquake', 7)
('ZoneC', 'Flood', 3)
('ZoneA', 'Flood', 9)
```

HIGH INTENSITY SENSOR EVENTS

```
-----
('ZoneA', 'Flood', 8)
('ZoneB', 'Earthquake', 7)
('ZoneA', 'Flood', 9)
```

SOCIAL MEDIA REPORT COUNT

```
-----
(('ZoneA', 'Flood'), 2)
(('ZoneB', 'Earthquake'), 1)
(('ZoneC', 'Flood'), 1)
```

FUSED EVIDENCE

```
-----
(('ZoneA', 'Flood'), 5)
(('ZoneB', 'Earthquake'), 3)
...
```

DISASTER ALERTS

```
-----
ALERT: ('ZoneA', 'Flood') | Evidence: 5
ALERT: ('ZoneB', 'Earthquake') | Evidence: 3
```

INTELLIGENT RESPONSE AGENT

```
-----
Agent response for ('ZoneA', 'Flood') : Iss early warning and recommend evacuation
```

2.

3.

```
TRANSACTION DATA
-----
('T1', 'U1', 500, 'India')
('T2', 'U2', 1200, 'India')
('T3', 'U1', 15000, 'Unknown')
...

SUSPICIOUS TRANSACTIONS
-----
('T3', 'U1', 15000, 'Unknown')
('T5', 'U2', 25000, 'Unknown')
('T7', 'U1', 18000, 'Unknown')

SUSPICIOUS TRANSACTIONS PER USER
-----
('U1', 2)
('U2', 1)

FRAUD DETECTION AGENT
-----
HIGH RISK USER: U1 -> Account requires investigation.
USER: U2 -> Transaction flagged for review.
```

4.

```
SUPPLY CHAIN ANALYSIS
-----
Laptop | Stock: 20 | Demand: 35 | Shortage: 15
Phone | Stock: 50 | Demand: 30 | Shortage: 0
Tablet | Stock: 15 | Demand: 25 | Shortage: 10
Monitor | Stock: 8 | Demand: 20 | Shortage: 12
Keyboard | Stock: 40 | Demand: 15 | Shortage: 0

INVENTORY AGENT
-----
Laptop -> Restock 15 units
Phone -> Inventory sufficient
Tablet -> Restock 10 units
Monitor -> Restock 12 units
Keyboard -> Inventory sufficient
```

5.

```
STUDENT LEARNING DATA
-----
('S1', 'Python', 85, 8)
('S1', 'Math', 55, 4)
('S1', 'AI', 70, 6)
...

AVERAGE STUDENT SCORE
-----
S1 Average: 70.0
S2 Average: 76.67
S3 Average: 53.33

WEAK TOPICS
-----
S1 -> Math | Score: 55
S2 -> Python | Score: 60
S3 -> Python | Score: 45
S3 -> AI | Score: 50

LEARNING AGENT RECOMMENDATIONS
-----
S1 | Weak Subject: Math | Recommendation: Revision videos + practice exercises
S3 | Weak Subject: Python | Recommendation: Beginner lessons + practice quizzes

AGENT FEEDBACK ADAPTATION
-----
S1 -> Continue current learning path
S2 -> Continue current learning path
S3 -> Agent recommends additional support
```